

ALGEBRAIC TOPOLOGY

ADAM KELLY – PART II

1. THE FUNDAMENTAL GROUP

1.1. Homotopy.

Definition 1.1 (Homotopy)

Let $f, g : X \rightarrow Y$ be maps. A **homotopy** from f to g is a map $H : X \times I \rightarrow Y$ such that $H(x, 0) = f(x)$, and $H(x, 1) = g(x)$.

If such a H exists, we say f is **homotopic** to g , and write $f \simeq h$ or $f \simeq_H g$.

Definition 1.2 (Homotopy Relative to a Subspace)

We say f is homotopic to g **relative to A** , written $f \simeq g \text{ rel } A$ if for all $a \in A \subseteq X$, we have $H(a, t) = f(a) = g(a)$.

Proposition 1.3

For spaces X, Y and $A \subseteq X$, ‘homotopic rel A ’ is an equivalence relation. In particular, when $A = \emptyset$, homotopy is an equivalence relation.

Proof. We check

- (1) *Reflexivity:* $f \simeq f$ since $H(x, t) = f(x)$ is a homotopy.
- (2) *Symmetry:* if $H(x, t)$ is a homotopy from f to g , then $H(x, 1 - t)$ is a homotopy from g to f .
- (3) *Transitivity:* Suppose $f, g, h : X \rightarrow Y$ and $f \simeq_H g \text{ rel } A, g \simeq_{H'} h \text{ rel } A$. We want to show that $f \simeq h \text{ rel } A$. The idea is to “glue” the two maps together.

We know how to continuously deform f to g , and from g to h . So we just do these one after another. We define $H'' : X \times I \rightarrow Y$ by

$$H''(x, t) = \begin{cases} H(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ H'(x, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

It’s well-defined since $H(x, 1) = g(x) = H'(x, 0)$. This is also continuous by the gluing lemma. It is easy to check that H'' is a homotopy rel A . $\square$

Definition 1.4 (Homotopy Equivalence)

A map $f : X \rightarrow Y$ is a **homotopy equivalence** if there exists a $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$

and $g \circ f \simeq \text{id}_X$. We call g a **homotopy inverse** for f .

If a homotopy equivalence $f : X \rightarrow Y$ exists, we say X and Y are **homotopy equivalent**, and write $X \simeq Y$.

Lemma 1.5

Consider the spaces and maps

$$\begin{array}{ccccc} X & \xrightarrow{f_0} & Y & \xrightarrow{g_0} & Z \\ & \searrow f_1 & \swarrow g_1 & & \\ & & Y & \xrightarrow{g_1} & Z \end{array}$$

If $f_0 \simeq_H f_1$ and $g_0 \simeq_{H'} g_1$, then $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Proof. We will show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$, and will then be done since homotopy is an equivalence relation. We write down two homotopies.

- (1) Consider the composition

$$X \times I \xrightarrow{H} Y \xrightarrow{g_0} Z$$

It is easy to check that this is the first homotopy we need to show $g_0 \circ f_0 \simeq g_0 f_1$.

- (2) The following composition is a homotopy from $g_0 \circ f_1$ to $g_1 \circ f_1$:

$$X \times I \xrightarrow{f_1 \times \text{id}_I} Y \times I \xrightarrow{H'} Z$$

$\square$

Proposition 1.6

Homotopy equivalence of spaces is an equivalence relation.

Proof. Symmetry and reflexivity are trivial. To show transitivity, let $f : X \rightarrow Y$ and $h : Y \rightarrow Z$ be homotopy equivalences, and $g : Y \rightarrow X$ and $k : Z \rightarrow Y$ be their homotopy inverses. We will show that $h \circ f : X \rightarrow Z$ is a homotopy equivalence with homotopy inverse $g \circ k$. We have

$$\begin{aligned} (h \circ f) \circ (g \circ k) &= h \circ (f \circ g) \circ k \\ &\simeq h \circ \text{id}_Y \circ k = h \circ k \simeq \text{id}_Z. \end{aligned}$$

Similarly,

$$\begin{aligned} (g \circ k) \circ (h \circ f) &= g \circ (k \circ h) \circ f \\ &\simeq g \circ \text{id}_Y \circ f = g \circ f \simeq \text{id}_X, \end{aligned}$$

as required. $\square$

Definition 1.7 (Contractible)

If $X \simeq *$, the one point space, then X is **contractible**.

Definition 1.8 (Retraction)

Let $A \subseteq X$ be a subspace. A **retraction** $r : X \rightarrow A$ is a map such that $r \circ \iota = \text{id}_A$, where $\iota : A \hookrightarrow X$

is the inclusion. If such an r exists, we call A a **retract** of X .

The retraction r is a **deformation retraction** if $\iota \circ r \simeq \text{id}_X$. A deformation retraction is **strong** if we require this homotopy to be a homotopy rel A .

1.2. Paths.

Definition 1.9 (Path)

A **path** in a space X is a map $\gamma : I \rightarrow X$. If $\gamma(0) = x_0$ and $\gamma(1) = x_1$, we say γ is a path from x_0 to x_1 , and write $\gamma : x_0 \rightsquigarrow x_1$.

If $\gamma(0) = \gamma(1)$, we say γ is a **loop** (based at x_0).

Definition 1.10 (Concatenation and Inverses)

If we have two paths γ_1 from x_0 to x_1 and γ_2 from x_1 to x_2 , we define the **concatenation** to be

$$(\gamma_1 \cdot \gamma_2)(t) = \begin{cases} \gamma_1(2t) & 0 \leq t \leq \frac{1}{2} \\ \gamma_2(2t - 1) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

This is continuous by the gluing lemma.

The **inverse** of a path $\gamma : I \rightarrow X$ is defined by

$$\gamma^{-1}(t) = \gamma(1 - t).$$

The **constant path** at a point $x \in X$ is given by $c_x(t) = x$.

Definition 1.11 (Homotopy of Paths)

Paths $\gamma, \gamma' : I \rightarrow X$ are **homotopic as paths** if they are homotopic rel $\{0, 1\} \subset I$, i.e. the end points are fixed. We write $\gamma \simeq \gamma'$.

Lemma 1.12 (Group Operation is Well-Defined)

Let $\gamma_1, \gamma_2 : I \rightarrow X$ be paths with $\gamma_1(1) = \gamma_2(0)$. Then if $\gamma_1 \simeq \gamma'_1$ and $\gamma_2 \simeq \gamma'_2$, then $\gamma_1 \cdot \gamma_2 \simeq \gamma'_1 \cdot \gamma'_2$.

Proof. Suppose that $\gamma_1 \simeq_{H_1} \gamma'_1$ and $\gamma_2 \simeq_{H_2} \gamma'_2$. Then we can construct the homotopy

$$H(s, t) = \begin{cases} H_1(s, 2t) & 0 \leq t \leq \frac{1}{2} \\ H_2(s, 2t - 1) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

□

Lemma 1.13 (Indeed a Group Operation)

Let $\gamma_0 : x_0 \rightsquigarrow x_1$, $\gamma_1 : x_1 \rightsquigarrow x_2$, $\gamma_2 : x_2 \rightsquigarrow x_3$ be paths. Then

- (1) $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2 \simeq \gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$.
- (2) $\gamma_0 \cdot c_{x_1} \simeq \gamma_0 \simeq c_{x_0} \cdot \gamma_0$.
- (3) $\gamma_0 \cdot \gamma_0^{-1} \simeq c_{x_0}$ and $\gamma_0^{-1} \cdot \gamma_0 \simeq c_{x_1}$.

Proof. We first note that the composition of any path γ and an order-preserving surjection $\phi : I \rightarrow I$ is homotopic to the original path via $H(s, t) = \gamma(t\phi(s) + (1 - t)s)$. Such a ϕ is a **reparameterisation**.

- (1) Take ϕ to be the function

$$\phi(t) = \begin{cases} \frac{t}{2} & 0 \leq t \leq \frac{1}{2} \\ t - \frac{1}{4} & \frac{1}{2} \leq t \leq \frac{3}{4} \\ 2t - 1 & \frac{3}{4} \leq t \leq 1. \end{cases}$$

Then we have $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2 \simeq ((\gamma_0 \cdot \gamma_1) \cdot \gamma_2) \circ \phi = \gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$.

- (2) To see $\gamma_0 \cdot c_{x_1} \simeq \gamma_0$, reparameterise γ_0 with

$$t \mapsto \begin{cases} 2t & 0 \leq t \leq \frac{1}{2} \\ 1 & \frac{1}{2} \leq t \leq 1, \end{cases}$$

and the other side is analogous.

- (3) We have that $c_{x_0} \simeq \gamma_0 \cdot \gamma_0^{-1}$ via

$$H(s, t) = \begin{cases} \gamma_0(2s) & 0 \leq s \leq \frac{t}{2} \\ \gamma_0(t) & \frac{t}{2} \leq s \leq 1 - \frac{t}{2} \\ \gamma_0^{-1}(2s - 1) & 1 - \frac{t}{2} \leq s \leq 1. \end{cases}$$

□

1.3. The Fundamental Group. We can now define one of our central objects of study.

Definition 1.14 (Fundamental Group)

Let X be a space and $x_0 \in X$. The **fundamental group** of X (based at x_0), denoted $\pi_1(X, x_0)$, is the set of homotopy classes of loops in X based at x_0 .

The group operation is defined to be $[\gamma_0][\gamma_1] = [\gamma_0 \cdot \gamma_1]$, and $[\gamma]^{-1} = [\gamma^{-1}]$ with the identity as the constant path $e = [c_{x_0}]$.

Theorem 1.15

The fundamental group is a group.

Proof. Immediate from our previous lemmas. □

Definition 1.16 (Based Space)

A **based space** is a pair (X, x_0) of a space X and a point $x_0 \in X$, the **basepoint**. A **map of based spaces** $f : (X, x_0) \rightarrow (Y, y_0)$ is a continuous map $f : X \rightarrow Y$ such that $f(x_0) = y_0$. A **based homotopy** is a homotopy rel $\{x_0\}$.

Proposition 1.17 (π_1 is a Functor)

To a based map $f : (X, x_0) \rightarrow (Y, y_0)$, there is an associated function $f_* = \pi_1(f) : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$, defined by $[\gamma] \mapsto [f \circ \gamma]$. Moreover it satisfies

- (1) $\pi_1(f)$ is a homomorphism of groups.

- (2) If $f \simeq f'$, then $\pi_1(f) = \pi_1(f')$.
- (3) For any maps $(A, a) \xrightarrow{h} (B, b) \xrightarrow{k} (C, c)$, we have $\pi_1(k \circ h) = \pi_1(k) \circ \pi_1(h)$.
- (4) $\pi_1(\text{id}_X) = \text{id}_{\pi_1(X, x_0)}$.

We want to remove the basepoint dependence of the fundamental group. We will assume that X is path connected.

Proposition 1.18

A path $u : x_0 \rightsquigarrow x_1$ induces a group isomorphism $u_\# : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ by $[\gamma] \mapsto [u^{-1} \cdot \gamma \cdot u]$. This satisfies

- (1) If $u \simeq u'$, then $u_\# = u'_\#$.
- (2) $(c_{x_0})_\# = \text{id}_{\pi_1(X, x_0)}$.
- (3) If $v : x_1 \rightsquigarrow x_2$, then $(u \cdot v)_\# = v_\# \circ u_\#$.
- (4) If $f : X \rightarrow Y$ with $f(x_0) = y_0$ and $f(x_1) = y_1$, then $(f \circ u)_\# \circ f_* = f_* \circ u_\# : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_1)$.
- (5) If $x_1 = x_0$, then $u_\#$ is an automorphism of $\pi_1(X, x_0)$ given by conjugation by u .

So if x_0 and x_1 are in the same path connected component, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

Lemma 1.19

The following diagram commutes.

$$\begin{array}{ccc}
 & \pi_1(Y, f(x_0)) & \\
 f_* \nearrow & \downarrow u_\# & \searrow g_* \\
 \pi_1(X, x_0) & & \pi_1(Y, g(x_0))
 \end{array}$$

That is, $g_* = u_\# \circ f_*$.

Proof. Suppose we have a loop $\gamma : I \rightarrow X$ based at x_0 .

We need to check that

$$g_*([\gamma]) = u_\# \circ f_*([\gamma])$$

In other words, we want to show that

$$g \circ \gamma \simeq u^{-1} \cdot (f \circ \gamma) \cdot u$$

To prove this result, we want to build a homotopy.

Consider the composition:

$$F : I \times I \xrightarrow{\gamma \times \text{id}_I} X \times I \xrightarrow{H} Y$$

Our plan is to exhibit two homotopic paths ℓ^+ and ℓ^- in $I \times I$ such that

$$F \circ \ell^+ = g \circ \gamma, \quad F \circ \ell^- = u^{-1} \cdot (f \circ \gamma) \cdot u$$

This is in general a good strategy — X is a complicated and horrible space we don't understand. So to construct

a homotopy, we make ourselves work in a much nicer space $I \times I$.

Our ℓ^+ and ℓ^- are defined in a rather simple way. Precisely, ℓ^+ is the path $s \mapsto (s, 1)$, and ℓ^- is the concatenation of the paths $s \mapsto (0, 1 - s)$, $s \mapsto (s, 0)$ and $s \mapsto (1, s)$

Note that ℓ^+ and ℓ^- are homotopic as paths. If this is not obvious, we can manually check the homotopy

$$L(s, t) = t\ell^+(s) + (1 - t)\ell^-(s)$$

This works because $I \times I$ is convex. Hence $F \circ \ell^+ \simeq F \circ L \circ F \circ \ell^-$ as paths.

Now we check that the compositions $F \circ \ell^\pm$ are indeed what we want. We have

$$F \circ \ell^+(s) = H(\gamma(s), 1) = g \circ \gamma(s)$$

Similarly, we can show that

$$F \circ \ell^-(s) = u^{-1} \cdot (f \circ \gamma) \cdot u(s)$$

So done. $\square$

Theorem 1.20

If $f : X \rightarrow Y$ is a homotopy equivalence, and $x_0 \in X$, then the induced map $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ is an isomorphism.

Proof. Let $g : Y \rightarrow X$ be a homotopy inverse. So $f \circ g \simeq_H \text{id}_Y$ and $g \circ f \simeq_{H'} \text{id}_X$.

We have no guarantee that $g \circ f(x_0) = x_0$, but we know that our homotopy H' gives us $u' = H'(x_0, \cdot) : x_0 \rightsquigarrow g \circ f(x_0)$.

Applying our previous lemma with id_X for “ f ” and $g \circ f$ for “ g ”, we get

$$u'_\# \circ (\text{id}_X)_* = (g \circ f)_*$$

Using the properties of the $*$ operation, we get that

$$g_* \circ f_* = u'_\#$$

However, we know that $u'_\#$ is an isomorphism. So f_* is injective and g_* is surjective.

Doing it the other way round with $f \circ g$ instead of $g \circ f$, we know that g_* is injective and f_* is surjective. So both of them are isomorphisms. $\square$

Definition 1.21 (Simply Connected Space)

A space is **simply connected** if it is path connected and $\pi_1(X, x_0) \equiv 1$ for some (any) choice of $x_0 \in X$.

Lemma 1.22

A path-connected space X is simply connected if and only if for any $x_0, x_1 \in X$, there exists a unique homotopy class of paths $x_0 \rightsquigarrow x_1$.

Proof. Suppose X is simply connected, and let $u, v : x_0 \rightsquigarrow x_1$ be paths. Now note that $u \cdot v^{-1}$ is a loop based at x_0 , it is homotopic to the constant path, and $v^{-1} \cdot v$ is trivially homotopic to the constant path. So we have

$$u \simeq u \cdot v^{-1} \cdot v \simeq v$$

On the other hand, suppose there is a unique homotopy class of paths $x_0 \rightsquigarrow x_1$ for all $x_0, x_1 \in X$. Then in particular there is a unique homotopy class of loops based at x_0 . So $\pi_1(X, x_0)$ is trivial. $\square$